

数学练习 第 06 份 答案与解析

共 63 题 与《练习06》题号一一对应

1. 由高斯公式, $\operatorname{div} = \frac{\partial}{\partial x}[xf(z)] + \frac{\partial}{\partial y}[zf(x)] + \frac{\partial}{\partial z}[xf(y)] = f(z)$, 故

$$I(t) = \iiint_{\Omega} f(z) dV.$$

Ω 是球心 $(0, t, 0)$ 、半径 t 的球体, $t \rightarrow 0^+$ 时 $f(z) = f(0) + f'(0)z + o(z)$, 而 $\iiint_{\Omega} z dV = 0$ (形心的 z 坐标为 0),

$$\lim_{t \rightarrow 0^+} \frac{I(t)}{t^3} = \lim_{t \rightarrow 0^+} \frac{f(0) \cdot \frac{4}{3}\pi t^3 + o(t^3)}{t^3} = \frac{4\pi}{3}.$$

2. 由 $\lambda e^{-\lambda} = \frac{\lambda^2}{2} e^{-\lambda}$ 得 $\lambda = 2$, 故

$$P\{X > 1\} = 1 - e^{-2} - 2e^{-2} = 1 - 3e^{-2}.$$

3. 曲线与区域关于 $y = x$ 对称, 而 $(x - y)^3$ 在交换 x, y 时变号, 故 $\iint_D (x - y)^3 d\sigma = 0$, $I = S(D)$.

令 $y = tx$ ($t \geq 0$), 代入 $(x + y)^3 = xy$ 得 $x = \frac{t}{(1+t)^3}$, 且 $x dy - y dx = x^2 dt$,

$$S = \frac{1}{2} \oint (x dy - y dx) = \frac{1}{2} \int_0^{+\infty} \frac{t^2}{(1+t)^6} dt = \frac{1}{2} \cdot \frac{1}{30} = \frac{1}{60}.$$

故 $I = \frac{1}{60}$.

4. 设 X, Y 独立同服从 $U[0, 1]$, $Z = |X - Y|$. $F_Z(z) = 1 - (1 - z)^2$ ($0 \leq z \leq 1$), 故

$$f_Z(z) = \begin{cases} 2(1 - z), & 0 \leq z \leq 1, \\ 0, & \text{其他.} \end{cases}$$

5. 定义域: $x > 0$ 或 $x < -\frac{1}{2}$.

$x \rightarrow +\infty$: $\sqrt{4x^2 + x} = 2x + \frac{1}{4} + o(1)$, $\ln(2 + \frac{1}{x}) = \ln 2 + \frac{1}{2x} + o(\frac{1}{x})$,

$$y = 2x \ln 2 + 1 + \frac{\ln 2}{4} + o(1) \implies y = (2 \ln 2)x + 1 + \frac{\ln 2}{4};$$

$x \rightarrow -\infty$: 同理 $\sqrt{4x^2 + x} = -2x - \frac{1}{4} + o(1)$, 得 $y = -(2 \ln 2)x - 1 - \frac{\ln 2}{4}$;

$x \rightarrow (-\frac{1}{2})^-$: $\ln(2 + \frac{1}{x}) \rightarrow -\infty$ 而根式趋于 $\frac{1}{\sqrt{2}}$, 故 $x = -\frac{1}{2}$ 为铅直渐近线.

($x \rightarrow 0^+$ 时 $y \rightarrow 0$, 无渐近线.) 共 3 条.

6. 由 $a_n = \ln(a_n + e^{b_n})$ 得 $e^{b_n} = e^{a_n} - a_n$.

当 $a_n \rightarrow 0$ 时 $e^{b_n} = 1 + \frac{a_n^2}{2} + o(a_n^2)$, 即 $b_n \sim \frac{a_n^2}{2}$.

A、C、D 均正确 ($\sum a_n$ 收敛 $\implies a_n \rightarrow 0 \implies \sum b_n, \sum b_n^2$ 收敛).

B 错: 取 $a_n = n^{-0.9}$, 则 $\sum a_n$ 发散而 $b_n \sim \frac{1}{2}n^{-1.8}$, $\sum b_n$ 收敛. 选 B.

7. 由 $r(A) = s \leq n$ 与 $r(B) = n \leq s$ 得 $s = n$, 故 A, B, AB 都是 n 阶可逆矩阵.

A, B, C 中的分块矩阵秩均为 $2n$, 只有零解; 而 D 经列变换 $c_2 - c_1 B$ 得

$$\begin{pmatrix} A & AB \\ E & B \end{pmatrix} \rightarrow \begin{pmatrix} A & O \\ E & O \end{pmatrix}, \quad r = n < 2n,$$

故有非零解. 选 D .

8. $|\sin t|$ 以 π 为周期且一个周期上的积分为 2, 故

$$\lim_{x \rightarrow +\infty} \frac{1}{x} \int_0^x |\sin t| dt = \frac{2}{\pi}.$$

9. 置换矩阵是正交的 0-1 矩阵: $A^T = A^{-1}$ 仍是置换矩阵, ①②正确.

而 $A^* = |A|A^{-1} = \pm A^{-1}$, 当 $|A| = -1$ (奇数次互换) 时 A^* 的元素含 -1 , 不是置换矩阵, ③错;

$A^{-1} - A^* = (1 - |A|)A^{-1}$ 未必为零, ④错. 选 C .

10. (I) 方程化为 $y' + \frac{y}{x^2} = e^{1/x}$, 积分因子 $e^{-1/x}: (ye^{-1/x})' = 1, y = (x + C)e^{1/x}$;

由 $y(1) = 3e$ 得 $C = 2$, 即 $y = (x + 2)e^{\frac{1}{x}}$.

$x \rightarrow 0^+$ 时 $y \rightarrow +\infty$, 得铅直渐近线 $x = 0$; $x \rightarrow +\infty$ 时 $y = x + 3 + o(1)$, 得斜渐近线 $y = x + 3$.

(II) $y' = e^{\frac{1}{x}} \frac{(x-2)(x+1)}{x^2}$, 故 y 在 $(0, 2)$ 减、 $(2, +\infty)$ 增, 最小值 $y(2) = 4\sqrt{e}$, 且两端趋于 $+\infty$.

故 $k > 4\sqrt{e}$ 时有 2 个交点; $k = 4\sqrt{e}$ 时 1 个; $0 < k < 4\sqrt{e}$ 时无交点.

11. 取定球为 $x^2 + y^2 + z^2 = 2az$ (球心 $(0, 0, a)$ 、半径 a), 动球心取原点: $x^2 + y^2 + z^2 = R^2$.

动球在定球内的部分满足 $R^2 \leq 2az$, 即 $z \geq \frac{R^2}{2a}$, 是球冠, 面积

$$S(R) = 2\pi R \left(R - \frac{R^2}{2a} \right) = 2\pi R^2 - \frac{\pi R^3}{a}.$$

$S'(R) = \pi R \left(4 - \frac{3R}{a} \right) = 0 \Rightarrow R = \frac{4a}{3}$ (极大). 故 $R = \frac{4a}{3}$ 时面积最大, 最大值 $\frac{32\pi a^2}{27}$.

12. 设 $k_1\alpha_1 + k_2\alpha_2 + k_3\alpha_3 = 0$. 用 $A - E$ 作用 (注意 $(A - E)\alpha_1 = -3\alpha_1, (A - E)\alpha_2 = 0, (A - E)\alpha_3 = \alpha_2$):

$$-3k_1\alpha_1 + k_3\alpha_2 = 0.$$

再用 $A - E$ 作用得 $9k_1\alpha_1 = 0$, 故 $k_1 = 0$; 代回得 $k_3\alpha_2 = 0 \Rightarrow k_3 = 0$; 最后 $k_2\alpha_2 = 0 \Rightarrow k_2 = 0$.

故 $\alpha_1, \alpha_2, \alpha_3$ 线性无关.

13. $f'_x = -(1 + e^y)\sin x = 0 \Rightarrow x = k\pi$; $f'_y = e^y(\cos x - 1 - y) = 0 \Rightarrow y = \cos x - 1$.

驻点: $(2k\pi, 0)$ 与 $((2k+1)\pi, -2)$. 在驻点处 $f''_{xx} = -(1 + e^y)\cos x, f''_{xy} = -e^y\sin x = 0, f''_{yy} = -e^y$.

$(2k\pi, 0)$: $AC - B^2 = (-2)(-1) = 2 > 0, A < 0$, 取极大值 $f = 2$;

$((2k+1)\pi, -2)$: $AC - B^2 = -(1 + e^{-2})e^{-2} < 0$, 非极值.

14. 当 $t \rightarrow -1$ 时 $x, y \rightarrow \infty$.

$$k = \lim_{t \rightarrow -1} \frac{y}{x} = \lim_{t \rightarrow -1} t = -1, \quad b = \lim_{t \rightarrow -1} (y + x) = \lim_{t \rightarrow -1} \frac{t^2 + t}{1 + t^3} = \lim_{t \rightarrow -1} \frac{t}{1 - t + t^2} = -\frac{1}{3}.$$

故斜渐近线为 $y = -x - \frac{1}{3}$.

15. $(1+ax^2)^{\frac{1}{3}} - 1 \sim \frac{a}{3}x^2$, $\cos x - 1 \sim -\frac{x^2}{2}$, 等价要求 $\frac{a}{3} = -\frac{1}{2}$, 故 $a = -\frac{3}{2}$.

16. A, B, C 的特征值都是 $2, 2, 1$.

$r(A - 2E) = 1 \Rightarrow \lambda = 2$ 有 2 个线性无关特征向量, A 可对角化, $A \sim C$;

$r(B - 2E) = 2 \Rightarrow \lambda = 2$ 只有 1 个, B 不可对角化, $B \not\sim C$. 选 A.

17. $\{X_i^2\}$ 独立同分布且期望存在, 由辛钦大数定律 $\frac{1}{n} \sum X_i^2 \xrightarrow{P} E(X^2)$, 即

$$\lim_{n \rightarrow \infty} P\left\{\left|\frac{1}{n} \sum X_i^2 - E(X^2)\right| < \varepsilon\right\} = 1.$$

选 C.

18. (I) 视 $x = x(y)$: $\frac{dy}{dx} = \frac{1}{x'}$, $y'' = -\frac{x''}{(x')^3}$, 代入原方程并乘以 $(x')^3$ 得

$$x'' - 4x = e^{2y}.$$

(II) 齐次解 $C_1 e^{2y} + C_2 e^{-2y}$; 共振特解 $\frac{y}{4} e^{2y}$. 故通解

$$x = C_1 e^{2y} + C_2 e^{-2y} + \frac{y}{4} e^{2y}.$$

19. (I) 通项 $= \frac{\sqrt[n]{n}}{(1 + \frac{1}{n^2})^n} \rightarrow 1 \neq 0$, 发散.

(II) 比值 $\rightarrow a$: $0 < a < 1$ 收敛; $a > 1$ 发散; $a = 1$ 时为 $\sum \frac{1}{n^p}$, $p > 1$ 收敛, $p \leq 1$ 发散.

(III) $\int_0^n \sqrt{1+x^3} dx \sim \frac{2}{5} n^{\frac{5}{2}}$, 通项 $\sim \frac{5}{2} n^{-\frac{5}{2}}$, 收敛.

(IV) 裂项: 部分和 $= a - a^{\frac{1}{n+1}} \rightarrow a - 1$, 收敛 (和为 $a - 1$).

(V) $\frac{1}{n} - \ln(1 + \frac{1}{n}) \sim \frac{1}{2n^2}$, 收敛.

(VI) $\sqrt[n]{a} - \sqrt{1 + \frac{1}{n}} = \frac{\ln a - \frac{1}{2}}{n} + O(\frac{1}{n^2})$, 故 $a \neq \sqrt{e}$ 时发散, $a = \sqrt{e}$ 时收敛.

20. $f_X(x) = \int_0^{+\infty} x e^{-x(1+y)} dy = e^{-x} (x > 0)$, 即 $X \sim E(1)$, $EX = 1$. 故

$$P\{X > 2 | X > 1\} = \frac{e^{-2}}{e^{-1}} = e^{-1}$$

(X 为连续型, 不等号取 $>$ 或 $\geq$ 结果相同).

21. $y + 1 \geq \frac{2}{1+x} \iff x + y + xy \geq 1 \iff y \geq \frac{1-x}{1+x}$. 记 D_2 为曲线下方部分,

$$\iint_{D_2} \frac{2}{1+x} d\sigma = 2 \int_0^1 \frac{1-x}{(1+x)^2} dx = 2(1 - \ln 2),$$

$$\iint_D (y+1) d\sigma = \frac{1}{3} + \frac{\pi}{4}, \quad \iint_{D_2} (y+1) d\sigma = \int_0^1 \left[\frac{g^2}{2} + g \right] dx = \frac{1}{2} \quad \left(g = \frac{1-x}{1+x} \right).$$

故

$$I = \left(\frac{1}{3} + \frac{\pi}{4} - \frac{1}{2} \right) + 2(1 - \ln 2) = \frac{\pi}{4} + \frac{11}{6} - 2 \ln 2.$$

22. 矩估计用样本二阶原点矩代替总体二阶原点矩：

$$\widehat{E(X^2)} = \frac{1}{n} \sum_{i=1}^n X_i^2 = \overline{X}^2 + \frac{1}{n} \sum_{i=1}^n (X_i - \overline{X})^2.$$

选 A .

23. $\operatorname{div} \mathbf{A} = (x^2 + 2xy + 2xz) + (y^2 + 2yz) + z^2 = x^2 + y^2 + z^2 + 2(xy + xz + yz).$

立体关于 z 轴旋转对称, 含 xy, xz, yz 的积分均为 0, 故

$$\Phi = \iiint_{\Omega} (x^2 + y^2 + z^2) dV = \int_0^{2\pi} \int_0^{\frac{\pi}{3}} \int_0^{4 \cos \varphi} \rho^4 \sin \varphi d\rho d\varphi d\theta = 2\pi \cdot \frac{1024}{5} \cdot \frac{63}{384} = \frac{336\pi}{5}.$$

24. 用分块合同变换 (舒尔补) :

$$\begin{pmatrix} E & 0 \\ -\alpha^T & 1 \end{pmatrix} \begin{pmatrix} A + \alpha\alpha^T & \alpha \\ \alpha^T & 1 \end{pmatrix} \begin{pmatrix} E & -\alpha \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & 1 \end{pmatrix}.$$

$A = \begin{pmatrix} -1 & 1 \\ 1 & 1 \end{pmatrix}$ 的特征值为 $\pm\sqrt{2}$, 故正、负惯性指数分别为 2, 1, 规范形 $y_1^2 + y_2^2 - y_3^2$. 选 C .

25. 记 $u = (1, 2, 3)^T$, $v = (3, 2, 1)^T$, 则 $A = \begin{pmatrix} O & u \\ v^T & 0 \end{pmatrix}$, $v^T u = 10$,

$$A^2 = \begin{pmatrix} uv^T & 0 \\ 0 & 10 \end{pmatrix}, \quad A^3 = 10A.$$

故

$$A^n = \begin{cases} 10^{\frac{n-1}{2}} A, & n \text{ 为奇数,} \\ 10^{\frac{n}{2}-1} A^2, & n \text{ 为偶数,} \end{cases} \quad A^2 = \begin{pmatrix} 3 & 2 & 1 & 0 \\ 6 & 4 & 2 & 0 \\ 9 & 6 & 3 & 0 \\ 0 & 0 & 0 & 10 \end{pmatrix}.$$

26. $F = xy - z \ln y + e^{xz} - 1$, 在 $(0, 1, 1)$ 处

$$F'_x = y + ze^{xz} = 2 \neq 0, \quad F'_y = x - \frac{z}{y} = -1 \neq 0, \quad F'_z = -\ln y + xe^{xz} = 0.$$

故可确定 $x = x(y, z)$ 与 $y = y(x, z)$, 而不能 (由隐函数定理) 确定 $z = z(x, y)$. 选 C .

27. $(E - 2\alpha\alpha^T)\alpha = (1 - 2 \cdot 2)\alpha = -3\alpha$, 对 $\xi \perp \alpha$ 有 $(E - 2\alpha\alpha^T)\xi = \xi$,

故其特征值为 $-3, 1, \dots, 1$, 可逆且对称. 于是

$$B^T X = 0 \iff (E - 2\alpha\alpha^T)A^T X = 0 \iff A^T X = 0.$$

选 D .

28. 令 $u = t - x$, 条件化为 $\int_0^{f(x)} g(u) du = x^2 \ln(1+x)$. 由反函数积分公式

$$\int_0^{f(x)} f^{-1}(u) du = x f(x) - \int_0^x f(t) dt = x^2 \ln(1+x).$$

两边求导得 $x f'(x) = 2x \ln(1+x) + \frac{x^2}{1+x}$, 即 $f'(x) = 2 \ln(1+x) + \frac{x}{1+x}$,
 $f(x) = 2(1+x) \ln(1+x) - x - \ln(1+x) \quad (f(0) = 0),$

故 $f(1) = 4 \ln 2 - 1 - \ln 2 = 3 \ln 2 - 1$.

29. $|(\alpha + 2\beta) \times (3\alpha + \beta)| = |\alpha \times \beta - 6\alpha \times \beta| = 5|\alpha \times \beta| = 5 \sin \frac{\pi}{6} = \frac{5}{2}.$

30. (I) $|x| < 1$ 时 $\frac{|x|^{n^2}}{2^n} \rightarrow 0$ 且级数收敛; $|x| = 1$ 时为 $\sum \frac{(\pm 1)}{2^n}$ 收敛; $|x| > 1$ 时通项趋于 ∞ .
 收敛域 $[-1, 1]$.

(II) $\frac{x^2}{3} < 1$ 即 $|x| < \sqrt{3}$ 时收敛; $x = \pm\sqrt{3}$ 时通项 $\rightarrow \pm\sqrt{3} \neq 0$, 发散. 收敛域 $(-\sqrt{3}, \sqrt{3})$.

31. $P\{X = Y\} = \sum_{k=1}^{\infty} p^2(1-p)^{2k-2} = \frac{p^2}{1-(1-p)^2} = \frac{p}{2-p}.$

选 D.

32. (I) 依题意 $\int_p^x \sqrt{1+y'^2} dt = y'(x)$, 求导得 $\sqrt{1+y'^2} = y''$.

令 $p = y' : \frac{dp}{\sqrt{1+p^2}} = dx$, 由 $p(0) = 0$ 得 $p = \sinh x$, 再由 $y(0) = 1$ 得

$$y = \cosh x = \frac{e^x + e^{-x}}{2}.$$

(II) $s = \int_{\ln 2}^{\ln 3} \cosh x dx = \sinh(\ln 3) - \sinh(\ln 2) = \frac{4}{3} - \frac{3}{4} = \frac{7}{12}.$

33. 取 $f(x) = \frac{e^x}{x}$, $g(x) = \frac{1}{x}$ (在 $[a, b]$ 上 $x > 0$), 由柯西中值定理存在 $\xi \in (a, b)$ 使

$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{\frac{e^b}{b} - \frac{e^a}{a}}{\frac{1}{b} - \frac{1}{a}} = \frac{ae^b - be^a}{a - b},$$

而

$$\frac{f'(\xi)}{g'(\xi)} = \frac{e^\xi(\xi - 1)/\xi^2}{-1/\xi^2} = e^\xi(1 - \xi).$$

两者相等即得结论.

34. 补上 $y = 0$ 上的圆盘 $x^2 + z^2 \leq 4$ (外法向为 $-\mathbf{j}$, 该面上三项积分均为 0), 由高斯公式

$$\operatorname{div} = x^2 + z^2, \quad I = \iiint_V (x^2 + z^2) dV = \int_0^{2\pi} \int_0^2 r^2(4 - r^2)r dr d\theta = \frac{32\pi}{3}.$$

35. $Y = \max\{X, \frac{1}{X}\} \geq 1$. 当 $y \geq 1$ 时

$$F_Y(y) = P\left\{\frac{1}{y} \leq X \leq y\right\} = e^{-\frac{1}{y}} - e^{-y},$$

$y < 1$ 时 $F_Y(y) = 0$.

36. (I) $\frac{\partial f(x, x+y)}{\partial x} = f'_u + f'_v = (u+v)e^{u-v} \Big|_{u=x, v=x+y} = (2x+y)e^{-y}$.

(II) 固定 $w = v - u$, 则 $\frac{d}{ds}f(s, s+w) = (2s+w)e^{-w}$, 积分并用 $f(0, w) = 0$ 得

$$f(u, u+w) = e^{-w}(u^2 + wu) \implies f(u, v) = uv e^{u-v}.$$

驻点: $f'_u = v(1+u)e^{u-v} = 0$, $f'_v = u(1-v)e^{u-v} = 0 \implies (0, 0)$ 与 $(-1, 1)$.

$(0, 0)$ 处 $f \approx uv$ 不取极值; $(-1, 1)$ 处 $A = C = e^{-2} > 0$, $B = 0$, $AC - B^2 > 0$,

故取极小值 $f(-1, 1) = -e^{-2}$.

37. 两个累次积分合并后的区域为 $D = \{x \geq 0, y \geq 0, 1 \leq x+y \leq 2\}$, 关于 $y = x$ 对称, 故

$$I = \iint_D e^{(x+y)^2} (\sin^2 x + \cos^2 y) d\sigma = \frac{1}{2} \iint_D e^{(x+y)^2} \cdot 2 d\sigma = \iint_D e^{(x+y)^2} d\sigma.$$

再用 $\iint_D g(x+y) d\sigma = \int_1^2 u g(u) du$:

$$I = \int_1^2 u e^{u^2} du = \frac{e^4 - e}{2}.$$

38. $(A+E)C = O$ 且 $r(C) = 1: \lambda = -1$ 至少有 1 个线性无关特征向量;

$B(A^T - 2E) = O$ 即 $(A - 2E)B^T = O$ 且 $r(B^T) = 2: \lambda = 2$ 至少有 2 个.

故特征值为 $-1, 2, 2$ 且可对角化, $A \sim \text{diag}(-1, 2, 2)$. 选 D.

39. 记 $F(x) = \int_1^x t^2 \sin t dt$, 因 $t^2 \sin t$ 为奇函数, $F(1) = F(-1) = 0$, 且 $F'(\pm 1) \neq 0$.

$x = 1$: 分子含因子 $x(x-1)$, $f \rightarrow \frac{2e^{-1}}{\sin 1}$, 可去间断点;

$x = -1$: 分子含 $|x+1|$, 左右极限为 $\pm \frac{2e}{\sin 1}$, 跳跃间断点;

$x = 0$: $x \rightarrow 0^-$ 时 $e^{-1/x} \rightarrow +\infty$ 使 $f \rightarrow \infty$, 为第二类间断点.

故第一类间断点共 2 个. 选 C.

40. 注意 $f'(x) \sin y dx + f(x) \cos y dy = d[f(x) \sin y]$, 而在 $A(2, 2\pi)$ 与 $O(0, 0)$ 处 $\sin y = 0$, 故这部分积分为 0:

$$I = -\pi \int_L x dy.$$

以 $x = 1 + R \cos t$, $y = \pi + R \sin t$ ($R = \sqrt{1 + \pi^2}$) 参数化上半圆周 (t 由 t_0 到 $t_0 + \pi$):

$$\int_L x dy = \int_{t_0}^{t_0+\pi} (1 + R \cos t) R \cos t dt = -2\pi + \frac{\pi(1 + \pi^2)}{2},$$

$$\text{故 } I = -\pi \left[\frac{\pi(1 + \pi^2)}{2} - 2\pi \right] = \frac{\pi^2(3 - \pi^2)}{2}.$$

41. 令 $u = y^2$, 方程化为 $u' - \frac{u}{x+1} = -\frac{x}{x+1}$, 积分因子 $\frac{1}{x+1}$:

$$\left(\frac{u}{x+1}\right)' = -\frac{x}{(x+1)^2} \Rightarrow \frac{u}{x+1} = -\ln|x+1| - \frac{1}{x+1} + C,$$

即通解

$$y^2 = C(x+1) - (x+1)\ln|x+1| - 1.$$

42. (I) 坐标关系 $x = Py$ 中

$$P = \begin{pmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix}$$

即为由基 α 到基 β 的过渡矩阵 $((\beta_1, \beta_2, \beta_3) = (\alpha_1, \alpha_2, \alpha_3)P)$.

(II)

$$\beta_1 = \alpha_1 - \alpha_2 = (-1, -1, 2)^T, \quad \beta_2 = \alpha_2 - \alpha_3 = (-1, 2, -1)^T, \quad \beta_3 = \alpha_3 = (3, 1, 2)^T.$$

43. X, Y 独立同服从 $U[0, 3]$: $EX = \frac{3}{2}$, $EX^2 = 3$, $EX^4 = \frac{81}{5}$.

$$EU = 2EX^2 + 2(EX)^2 = \frac{21}{2}, \quad EV = 2EX^2 - 2(EX)^2 = \frac{3}{2},$$

$$E(UV) = E(X^2 - Y^2)^2 = 2EX^4 - 2(EX^2)^2 = \frac{72}{5},$$

$$\text{Cov}(U, V) = \frac{72}{5} - \frac{21}{2} \cdot \frac{3}{2} = -\frac{27}{20}.$$

44. 当 $x \in [-3, -\frac{5}{2}]$ 时 $-x-2 \in [\frac{1}{2}, 1]$, 由周期性与偶性

$$S(x) = f(x) = f(-x-2) = 2 - 2(-x-2) = 2x+6 \quad \left(x \in \left[-3, -\frac{5}{2}\right]\right),$$

而 $x = -\frac{5}{2}$ 是 f 的跳跃点, $S(-\frac{5}{2}) = \frac{1+\frac{1}{2}}{2} = \frac{3}{4}$.

45. $A = 2E - J$ (J 为 4 阶全 1 矩阵), $J^2 = 4J$, 故

$$A^2 = 4E - 4J + J^2 = 4E.$$

于是

$$A^n = \begin{cases} 2^n E, & n \text{ 为偶数,} \\ 2^{n-1} A, & n \text{ 为奇数.} \end{cases}$$

46. $BA^T = O$ 且 $A \neq O$ 得 $|B| = a^2 - a - 6 = 0$, 由 $a > 0$ 得 $a = 3$, 此时 $r(B) = 2$.

A 的各行都属于 $Bx = 0$ 的解空间, 该空间由 $(-11, 6, 1)^T$ 张成, 故 $r(A) = 1$,

$AX = 0$ 即 $-11x_1 + 6x_2 + x_3 = 0$. 验证 $(1, 2, -1)^T, (0, -1, 6)^T$ 均为其解且线性无关. 选 B.

47. 二次型矩阵为 $\begin{pmatrix} 1 & 1 & 0 \\ 1 & k+2 & 0 \\ 0 & 0 & k \end{pmatrix}$, 椭球面要求其正定:

$$1 > 0, \quad k+1 > 0, \quad k(k+1) > 0 \Rightarrow k > 0.$$

选 A.

48. 记 $\varphi(x) = \int_0^x (e^{-t^2} + \sin t^2) dt$, 则 $f(x, \varphi(x)) \equiv 0$, $\varphi(0) = 0$, $\varphi'(0) = 1$.

求导得 $f'_x + f'_y \varphi' = 0$, 在 $(0, 0)$ 处 $1 + f'_y = 0$, $f'_y(0, 0) = -1$. 故

$$dz|_{(0,0)} = dx - dy.$$

49. 上侧时 $dS = (-z'_x, -z'_y, 1) dx dy = (-\frac{x}{r}, -\frac{y}{r}, 1) dx dy$ ($r = \sqrt{x^2 + y^2} = z$),

$$I = \iint_{1 \leq r \leq 2} \left[-x^2 r - \frac{y^3}{r} + rx \right] d\sigma = \int_0^{2\pi} \int_1^2 [-r^4 \cos^2 \theta - r^3 \sin^3 \theta + r^3 \cos \theta] dr d\theta = -\frac{31\pi}{5}.$$

50. $P(A) = 0.7$, $P(AB) = P(A) - P(A - B) = 0.2$, $P(A \cup \overline{B}) = P(A) + P(\overline{B}) - P(A\overline{B}) = 0.7 + 0.6 - 0.5 = 0.8$,

$$P(B | A \cup \overline{B}) = \frac{P(AB)}{P(A \cup \overline{B})} = \frac{0.2}{0.8} = \frac{1}{4}.$$

51. $x \rightarrow +\infty$ 时 $\frac{\ln x}{x^a} \rightarrow 0$ (需 $a > 0$), 此时被积函数 $\sim \frac{\ln x}{x^a}$,

而 $\int_2^{+\infty} \frac{\ln x}{x^a} dx$ 收敛当且仅当 $a > 1$; $a \leq 0$ 时被积函数不趋于 0, 发散. 选 C.

52. 由 $\frac{\partial}{\partial y}[-6yf] = \frac{\partial}{\partial x}[x^2 f' - 4xf]$ 得欧拉方程

$$x^2 f'' - 2xf' + 2f = 0 \implies f = C_1 x + C_2 x^2,$$

由 $f(1) = 1, f'(1) = 2$ 得 $C_1 = 0, C_2 = 1$, 即 $f(x) = x^2$. 此时

$$du = -6x^2 y dx - 2x^3 dy \implies u(x, y) = -2x^3 y + C.$$

53. $P(B) = P(A) = \frac{1}{2}$ 给出 $\sigma = \mu$, 于是 $B = A$, 且 $C = \{X > \mu + \sigma\} \subset A$.

因此只要 A 发生就至少有两个事件发生, "至多有一个发生"即 A 不发生:

$$P = P(\overline{A}) = \frac{1}{2}.$$

54. $f(x) = x^2 \sum_{n \geq 1} a_n x^{n-1}$, 由在 $[0, 1]$ 上收敛知 $h(x) = \sum a_n x^{n-1}$ 在 $[0, 1]$ 上有界, $|h| \leq M$.

故 $|f(\frac{1}{n})| \leq \frac{M}{n^2}$, 级数 $\sum (-1)^{n-1} f(\frac{1}{n})$ 绝对收敛. 选 C.

55. $\sin(xy)^2 \sim (xy)^2$,

$$\int_0^t dx \int_x^t x^2 y^2 dy = \int_0^t x^2 \cdot \frac{t^3 - x^3}{3} dx = \frac{t^6}{18},$$

故极限为 $\frac{1}{18}$.

56. (按 $x \rightarrow 0$ 理解, 原题 " $x \rightarrow \infty$ " 存疑.)

$x \rightarrow 0^+$: $\frac{\ln(1+e^{2/x})}{\ln(1+e^{1/x})} \rightarrow 2$, $[x] = 0$, $f \rightarrow 2$;

$x \rightarrow 0^-$: 该比值 $\rightarrow 0$, $[x] = -1$, $f \rightarrow -a$.

极限存在要求 $-a = 2$, 故 $a = -2$, $b = 2$.

(若严格按 $x \rightarrow \infty$, 则必须 $a = 0$, 此时 $b = 1$.)

57. $\alpha = P\left\{\overline{X} > \mu_0 + \frac{1.96}{\sqrt{n}} \mid \mu = \mu_0\right\} = P\{Z > 1.96\} = 1 - \Phi(1.96).$

选 C.

58. $\mathbf{F} = \frac{\mathbf{r}}{|\mathbf{r}|^3}$ 在原点外散度为零, 而 S 是包围原点的闭曲面,

挖去小球面并用高斯公式得

$$I = \oiint_{|\mathbf{r}|=\varepsilon} \frac{\mathbf{r} \cdot \mathbf{n}}{\varepsilon^3} dS = \frac{1}{\varepsilon^2} \cdot 4\pi\varepsilon^2 = 4\pi.$$

59. $F(0) = 0$,

$$F'_+(0) = \lim_{x \rightarrow 0^+} f(x) \frac{\sin x}{x} = \lim_{x \rightarrow 0^+} f(x), \quad F'_-(0) = - \lim_{x \rightarrow 0^-} f(x),$$

故可导 $\iff$ 两个单侧极限存在且 $\lim_{x \rightarrow 0^-} f(x) = - \lim_{x \rightarrow 0^+} f(x)$. 选 D.

60. 由 $x_1 = \frac{y_1}{2}$, $x_2 = \frac{y_2}{3}$, 雅可比行列式为 $\frac{1}{6}$, 故

$$f_2(y_1, y_2) = \frac{1}{6} f_1\left(\frac{y_1}{2}, \frac{y_2}{3}\right).$$

选 D.

61. 由 $f(tx, ty) = t^2 f(x, y)$ (二次齐次) 及欧拉公式 $xf'_x + yf'_y = 2f$, 在 $(1, -2)$ 处

$$4 - 2f'_y(1, -2) = 2 \cdot 2 = 4 \implies f'_y(1, -2) = 0.$$

切平面: $z - 2 = 4(x - 1) + 0 \cdot (y + 2)$, 即 $4x - z - 2 = 0$.

62. 由 $A^3\alpha = 3A\alpha - 2A^2\alpha$ 得 $AP = PB$, 其中

$$B = \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 3 \\ 0 & 1 & -2 \end{pmatrix},$$

故 $A \sim B$,

$$|A + E| = |B + E| = \begin{vmatrix} 1 & 0 & 0 \\ 1 & 1 & 3 \\ 0 & 1 & -1 \end{vmatrix} = -4.$$

63. 在 $x = b$ 处连续且可导:

$$a\sqrt{b} = \ln b, \quad \frac{a}{2\sqrt{b}} = \frac{1}{b} \implies a = \frac{2}{\sqrt{b}},$$

代入第一式得 $2 = \ln b$, $b = e^2$, $a = \frac{2}{e}$. 选 A.